Introduction to Operations Research, Deterministic Models

Lecture 11: Dynamic Programming

Department of Industrial Engineering, Tsinghua University

Instructor: Junyi Liu

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Recap: characteristics of DP

- The problem can be divided into **stages**, with a policy decision required at each stage. Each stage has a number of **states** associated with the beginning of that stage.
- The effect of the policy decision at each stage is to **transform the current state to a state** associated with the beginning of the next stage.
- Given the current state, an optimal policy for the remaining stages is independent of the policy decisions adopted in previous stages. This is the **principle of optimality** for DP.
- **A forward recursive relationship** identifies the optimal policy for stage n, given that the optimal policy for stage $n+1$ is available.
- When we use this recursive relationship, the solution procedure **starts at the end and moves backward stage by stage** —each time finding the optimal policy for that stage—until it finds the optimal policy starting at the initial stage.
- Alternatively, with **a backward recursive process**, the solution procedure moves forward stage by stage, where the first to be solved is the initial stage of the problem and problems are solved moving forward one stage at a time, until all stages are included.

Multi-stage inventory control problem

- Corporation A must determine how many products should be produced during each of the next 7 days. The demand during each of the next 7 day is: $(40, 60, 75, 25, 40, 60, 70)$. Corporation A must meet demands on time.

- At the beginning of the first day, Corporation A has an inventory of 10 products. At the beginning of each day, corporation A must decide how many products should be produced during that day. For simplicity, we assume that products manufactured during a day can be used to meet demand for that day and afterwards.

- During each day, corporation A can produce up to 40 products with regular-time labor at a total cost of $\$400$ per product. By having employees work overtime during a day, corporation A can produce additional products with overtime labor at a total cost of $\$450$ per product.

- At the end of each day (after production has occurred and the current day's demand has been satisfied), a holding cost of $\$20$ per product is incurred. Use linear programming to determine a production schedule to minimize the sum of production and inventory costs during the next 7 days.

Identify the state, stage, and decision.

Multi-stage inventory control problem

Stage: each of the next 7 days

State: inventory level at the beginning of each day

Decision: units of production

Recursion: let $f_t^*(y)$ represent the minimum cost of meeting demands for the remaining days $t, t+1, \ldots$ with the inventory level y at the beginning of day t.

$$f_t^*(y) = \min_{x \geq 0, y+x-d_t \geq 0} 20(y + x - d_t) + c(x) + f_{t+1}^*(y + x - d_t)$$

where d_t represents the demand for day t, and $c(x)$ represent the production costs for producing x products, i.e. $c(x) = \begin{cases} 400x & \text{if } x \leq 40 \\ 450x & \text{if } x > 40 \end{cases}$

Resource allocation problem

Resource allocation problem

Finco has $\$6000$ to invest, and three investments are available. If d_j dollars (in thousands) are invested in investment j, then a net present value of $r_j(d_j)$ is obtained:

$$r_1(d_1) = \begin{cases} 7d_1 + 2 & \text{if } d_1 > 0 \\ 0 & \text{if } d_1 = 0 \end{cases}$$

$$r_2(d_2) = \begin{cases} 3d_2 + 7 & \text{if } d_2 > 0 \\ 0 & \text{if } d_2 = 0 \end{cases}$$

$$r_3(d_3) = \begin{cases} 4d_3 + 5 & \text{if } d_3 > 0 \\ 0 & \text{if } d_2 = 0 \end{cases}$$

The amount placed in each investment must be an exact multiple of $\$1000$. To maximize the net present value obtained from the investments, how should Finco allocate the $\$6000$?

How to define the stages?

the stage should be chosen so that the problem is easy to solve when one stage remains.

Resource allocation problem

Stage: at stage t, a amount of money must be allocated to investments t.

Decision: $x_t =$ amount of money allocated to investments t.

State: the amount of money (in thousands) available at the beginning of stage t for the remaining investments $t, t+1, \ldots, 3$. The possible states are 0,1,2,3,4,5,6.

Recursion: let $f_t^*(d_t)$ represent the maximum net present value that can be obtained by investing d_t thousand dollars in investments $t, t+1, \ldots, 3$.

$$f_t^*(d_t) = \max_{0 \le x_t \le d_t, x_t \text{is integer}} \{r_t(x_t) + f_{t+1}^*(d_t - x_t)\}$$

$x_t^*(d_t)$ denote the optimal solution in the above maximization problem, which means the optimal amount of money allocated to investment t given d_t amount of money available for the remaining stages.

Resource allocation problem

Recursion: let $f_t^*(d_t)$ represent the maximum net present value that can be obtained by investing d_t thousand dollars in investments $t, t+1, \ldots, 3$.

$$f_t^*(d_t) = \max_{0 \le x_t \le d_t, x_t \text{is integer}} \{r_t(x_t) + f_{t+1}^*(d_t - x_t)\}$$

$x_t^*(d_t)$ denote the optimal solution, which means the optimal amount of money allocated to investment t given d_t amount of money available for the remaining stages.

Stage 3: calculate $f_3^*(d_3) = 4d_3 + 5$ with $x_t^*(d_3) = d_3$ for $d_3 = 1, \ldots, 6$, and $f_3^*(0) = 0$ with $x_t^*(0) = 0$

Stage 2: calculate $f_2^*(d_2) = \max_{0 \le x_2 \le d_2} r_2(x_2) + f_3^*(d_2 - x_2)$ for $d_2 = 0, \ldots, 6$.

Stage 1: calculate $f_1^*(6) = \max_{0 \le x_1 \le 6} r_1(x_1) + f_2^*(6 - x_1)$.

Recall: capital budgeting problem (application of LP)

Star Oil Company is considering five different investment opportunities. The cash outflows in time 0 and time 1, and NPV (in millions of dollars) are given in the following table.

	Inv 1	Inv 2	Inv 3	Inv 4	Inv 5
Time 0 cash outflow	11	53	5	5	29
Time 1 cash outflow	3	6	5	1	34
NPV	13	16	16	14	39

Star Oil Company has \$40 millions available for investments now (time 0); it estimates that one year from now (time 1) \$20 million will be available for investment.

Star Oil Company may purchase any fraction of the investment, and suppose that the cash outflows and NPV are adjusted proportionally.

The company wants to maximize the NPV that can be obtained by investing the five investments. Formulate an LP to help achieve the goal.

Recall: capital investment problem (application of BIP)

There are four investments and **each investment must be either fully invested or not invested.**

- Investment 1: Invest: 5,000; Return: 16,000
- Investment 2: Invest: 7,000; Return: 22,000
- Investment 3: Invest: 4,000; Return: 12,000
- Investment 4: Invest: 3,000; Return: 8,000

With the total budget of 14,000, the goal is to maximize the total return from all investments.

Decision variables: $x_i \in \{0, 1\}$ represents whether invest or not.

BIP problem:

$$\begin{aligned} \text{maximize} \quad & 16x_1 + 22x_2 + 12x_3 + 8x_4 \\ \text{subject to} \quad & 5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14, \\ & x_i \in \{0, 1\}, \forall 1 \leq i \leq 4. \end{aligned}$$

Generalized Resource Allocation Problem

Suppose we have w (a nonnegative integer) units of a resource available and T activities to which the resource can be allocated.

If activity t is implemented at a level x_t (a nonnegative integer), then $g_t(x_t)$ (a nonnegative integer) units of the resource are used by activity t, and a benefit $r_t(x_t)$ is obtained.

The problem is to determine the allocation of resources that maximizes total benefit subject to the limited resource availability.

$$\begin{aligned} \text{maximize} \quad & \sum_{t=1}^{T} r_t(x_t) \\ \text{s.t.} \quad & \sum_{t=1}^{T} g_t(x_t) \leq w \\ & x_t \in \mathcal{X},\ \forall t = 1, \ldots, T \end{aligned}$$

where $\mathcal{X}$ could be $\{0, 1\}$, $\{0, 1, 2, \ldots\}$, or $\{0, 1, 2, \ldots, C\}$.

Generalized Resource Allocation Problem

Interpretations of $r_t(x_t)$, $g_t(x_t)$ and w

Interpretation of $r_t(x_t)$	Interpretation of $g_t(x_t)$	Interpretation of w
Benefit from placing x_t of item t in a knapsack	Weight of x_t of item t	Maximal weight that knapsack can hold
Grade obtained in course t if one study course t for x_t hours	Number of hours spent studying course t	Total number of study hours available
Sales in region t if x_t sales representative are assigned to region t	Cost of assigning x_t sales reps in region t	Total sales force budget

Generalized Resource Allocation Problem

$$\begin{aligned} \text{maximize} \quad & \sum_{t=1}^{T} r_t(x_t) \\ \text{s.t.} \quad & \sum_{t=1}^{T} g_t(x_t) \leq w \\ & x_t \in \mathcal{X},\ \forall t = 1, \ldots, T \end{aligned}$$

where $\mathcal{X}$ could be $\{0, 1\}$, $\{0, 1, 2, \ldots\}$, or $\{0, 1, 2, \ldots, C\}$.

DP formulation:

Stage: at stage t, some amount of the resource must be allocated to activity t.
State: the amount of resource available at any stage t. The possible states are?
Recursion: $f_t^*(d_t)$ to be the maximum benefit that can be obtained from activities $t, t+1, \ldots T$ if d_t units of the resource are available.

$$f_t^*(d_t) = \max_{x_t \geq 0, g_t(x_t) \leq d_t, x_t \text{ is integer}} \{r_t(x_t) + f_{t+1}^*(d_t - g_t(x_t))\}$$

and $x_t^*(d_t)$ denote the optimal solution in the above maximization problem.

General discussion of DP

General discussion of DP

Suppose we have a multi-stage decision process.

- $N =$ number of stages
- $n =$ label for current state n
- $x_n =$ decision variable for stage n
- $s_n =$ the state at the beginning of stage n with the remaining stages $n, n+1, \ldots, N$ to go
- $f_n(x_n, s_n)$ represent the return (or cost) for a particular stage n
- $D_n(s_n)$ represent the set of feasible decisions which depends upon the state s_n at stage n.

Assume:

- the state s_n of the system with $N - n$ stages to go is a full description of the system for decision-making purposes and that knowledge of prior states is unnecessary.
- With the decision variable x_n and state s_n at stage n, the state for stage $n+1$ is $s_{n+1} = t_n(x_n, s_n)$ where t_n is referred to as **the transition function**.

General discussion of DP

Objective: maximize the sum of the return functions (or minimize the sum of cost functions) over all stages of the decision process;

Constraints:

the decision chosen for each stage belong to some set $D_n(s_n)$
the transitions from state to state follows the transition function

$$
\begin{aligned}
f_n^*(s_n) = \quad & \max \big(f_n(x_n, s_n) + f_{n+1}(x_{n+1}, s_{n+1}) + \ldots + f_N(x_N, s_N) \big) \\
& \text{s.t.} \quad s_{m+1} = t_m(x_m, s_m), \quad \forall m = n, \ldots, N-1. \\
& \qquad\quad x_m \in D_m(s_m), \quad \forall m = n, \ldots, N.
\end{aligned}
$$

How this is related to DP formulation?

General discussion of DP

$$
\begin{aligned}
f_n^*(s_n) = \quad & \max \left(f_n(x_n, s_n) + f_{n+1}(x_{n+1}, s_{n+1}) + \ldots + f_N(x_N, s_N) \right) \\
& \text{s.t.} \quad s_{m+1} = t_m(x_m, s_m), \quad \forall m = n, \ldots, N-1. \\
& \qquad\quad x_m \in D_m(s_m), \quad \forall m = n, \ldots, N.
\end{aligned}
$$

$\Longleftrightarrow$

$$
\begin{aligned}
f_n^*(s_n) \quad = \quad & \max \Big[f_n(x_n, s_n) + \qquad \max \left(f_{n+1}(x_{n+1}, s_{n+1}) + \ldots + f_N(x_N, s_N) \right) \Big] \\
& \text{s.t.} \quad x_n \in D_n(s_n) \qquad\qquad \text{s.t.} \quad s_{m+1} = t_m(x_m, s_m), \quad \forall m = n+1, \ldots, N-1 \\
& \qquad\quad s_{n+1} = t_n(x_n, s_n) \qquad\qquad x_m \in D_m(s_m), \quad \forall m = n+1, \ldots, N \\
= \quad & \max \Big[f_n(x_n, s_n) + \qquad f_{n+1}^*(t_n(x_n, s_n)) \Big] \\
& \text{s.t.} \quad x_n \in D_n(s_n)
\end{aligned}
$$

Principle of optimality

$$\boxed{f_n^*(s_n) = \max_{x_n \in D_n(s_n)} \Big[f_n(x_n, s_n) \ + f_{n+1}^*(t_n(x_n, s_n)) \Big]}$$

This principle says that an optimal sequence of decisions for a multistage problem has the property that, regardless of the current decision x_n and current state s_n, all subsequent decisions must be optimal, given the state $s_{n+1} = t_n(x_n, s_n)$ resulting from the current decision.

Relating to resource allocation problem:

Stage: at stage t, some amount of the resource must be allocated to activity t.
State: the amount of resource d_t available at any stage t.
Recursion: $f_t^*(d_t)$ to be the maximum benefit that can be obtained from activities $t, t+1, \ldots T$ if d_t units of the resource are available.

$$f_t^*(d_t) = \max_{x_t \geq 0, g_t(x_t) \leq d_t, x_t \text{ is integer}} \{r_t(x_t) + f_{t+1}^*(d_t - g_t(x_t))\}$$

General discussion of DP

$$\boxed{f_n^*(s_n) = \max_{x_n \in D_n(s_n)} \left[f_n(x_n, s_n) \ + f_{n+1}^*(t_n(x_n, s_n)) \right]}$$

• The stage-return functions or the set of permissible decisions at each stage could be various kinds.

• The decisions could be discrete, continuous, or mixtures of the two, as long as the recursive relationship be solvable for the optimal solution at each stage.

• The optimization problem that is defined at each stage could lead to the application of a wide variety of techniques, i.e., linear programming, network theory, integer programming, and so forth, depending on the nature of the transition function, the constraint set D_n, and the form of the function to be optimized.

• The recursion holds with state space, i.e., the set of possible states, being either discrete or continuous. If the state space is finite, then we could solve the DP by enumerating all possible states. What if the state space is infinite?

Continuous state-space problem

Continuous state-space problem

Example: Suppose that some governmental agency is attempting to to determine which programs should receive funding for the next fiscal year. The agency has managed to put together the amount of **expected benefit, expected cost, and Benefit/Cost ratio** in millions as follows.

program	expected benefit	expected cost	Benefit/Cost ratio
A	59.2	2.8	21.1
B	31.4	1.7	18.4
C	15.7	1.0	15.7
D	30.0	3.2	9.4
...	...	...	...
Total	364.2	62.5	

Either a program will be funded fully or it will not be considered for funding. Suppose that the agency has a budget of \$34 million from the state legislature. How can the agency make the most effective use of its funds at either possible budget level?

It is a type of resource allocation problem, with a special feature that the amount of resource are not integer-valued.

Let b_j denote the benefit of the jth program and c_j denote the cost of that program.
$f_n^*(B)$ = Maximum total benefit obtainable, choosing from the first n programs, with budget limitation B.

- Clearly, $f_0^*(B_0) = 0$, for $B_0 \geq 0$
- **Backward recursion:** $f_n^*(B_n) = \max\limits_{x_n \in \{0,1\}, c_n x_n \leq B_n} \{b_n x_n + f_{n-1}^*(B_n - c_n x_n)\}$

At stage 1,

- if $2.8 \leq B_1$, then $f_1^*(B_1) = 59.2$ and $x_1^* = 1$
- if $0 \leq B_1 < 2.8$, then $f_1^*(B_1) = 0$ and $x_1^* = 0$

At stage 2,

- if $4.5 \leq B_2$, then $f_2^*(B_2) = 59.2 + 31.4 = 90.6$ and $x_2^* = 1$
- if $2.8 \leq B_2 < 4.5$, then $f_2^*(B_2) = 59.2$ and $x_2^* = 0$
- if $1.7 \leq B_2 < 2.8$, then $f_2^*(B_2) = 31.4$ and $x_2^* = 1$
- if $0 \leq B_2 < 1.7$, then $f_2^*(B_2) = 0$ and $x_2^* = 0$

At stage 3, we can compute $f_3^*(B_3)$ for the following cases

- if $5.5 \leq B_3$,
- if $4.5 \leq B_3 < 5.5$,
- if $3.8 \leq B_3 < 4.5$,
- if $2.8 \leq B_3 < 3.8$,
- if $2.7 \leq B_3 < 2.8$,
- if $1.7 \leq B_3 < 2.7$,
- if $1.0 \leq B_3 < 1.7$,
- if $0 \leq B_3 < 1.0$.

Although this example has a continuous state space, a finite number of ranges for computing optimal benefit can be constructed because of the zero–one nature of the decision variables.

All breaks in the range of the state space either are the breaks from the previous stage, or they result from adding the cost of the new program to the breaks in the previous range.

We can recursively calculate the optimal-value function for succeeding stages, but the number of ranges that need to be reported rapidly becomes rather large.

Continuous state-space problem

It is not a general property of continuous state space problems to break the continuous space into fnite ranges, and in most cases such ranges cannot be determined.

Usually, **continuous state space problems are converted into discrete state problems** by defining an appropriate grid on the continuous state space. The optimal-value function is then computed only for the points on the grid.

For our cost/benefit example, the total budget must be between 0 and \$62.5 M, which provides a natual range on the state space, whose upper bound could be further tightened by the sum of the budgets of the first n programs.

An appropriate grid would consist of increments of \$0.1 M over the limits of the range at each stage, since this is the accuracy with which the program costs have been estimated.

Dynamic programming under uncertainty

Dynamic programming under uncertainty

In a deterministic dynamic-programming process: given the current state, a specific decision leads **with complete certainty** to a particular state at the next stage.

In a dynamic programming process under uncertainty: given the current state, a specific decision does not determine the state of the system at the next stage; this decision may not even determine the return for the current stage.

Process: Given the state of the system s_n at stage n and the current decision x_n,

• an uncertain event occurs which is determined by a random variable $\tilde{e}_n$ whose outcome e_n is not under the control of the decision maker.

• the stage return function $f_n(x_n, s_n, \tilde{e}_n)$ may depend on this random variable.

• the transition function $t_n(x_n, s_n, \tilde{e}_n)$ may also depend on this random variable; that is, the state for stage $n+1$ is $s_{n+1} = t_n(x_n, s_n, \tilde{e}_n)$.

• the outcomes e_n of the random variable $\tilde{e}_n$ are governed by a probability distribution, $p_n(e_n|x_n, s_n)$.

Dynamic programming under uncertainty

With the random stage return function $f_n(x_n, s_n, \tilde{e}_n)$ and the random transition function $t_n(x_n, s_n, \tilde{e}_n)$, we must optimize **the expected return** over the stages of the problem. In this situation, backward induction can be applied to determine the optimal strategy.

$$\boxed{f_n^*(s_n) = \max_{x_n \in D_n(s_n)} \left(\mathbf{E}\left[f_n(x_n, s_n, \tilde{e}_n) \ + f_{n+1}^*(t_n(x_n, s_n, \tilde{e}_n)) \right] \right)}$$

Application: inventory control

A manager is in charge of the replenishment decisions during the next two months for the inventory of a fairly expensive item. The production cost of the item is \$1000/unit, and its selling price is \$2000/unit.

There is an inventory-carrying cost of \$100/unit per month on each unit left over at the end of the month. We assume there is no setup cost associated with running a production order, any amount produced during a given month is available to satisfy the demand during that month.

At the present time, there is no inventory on hand. Any inventory left at the end of the next two months has to be disposed of at a salvage value of \$500/unit.

The demand for the item is uncertain, with its probability distribution identical for each of the coming two months. The probability distribution of the demand is as follows:

demand	probability
0	0.25
1	0.4
2	0.2
3	0.15

Application: inventory control

Question: how many units to produce during the first month and, depending on the actual demand in the first month, how many units to produce during the second month to minimize the total cost.

Observation:
- since demand is uncertain, the inventory at the end of each month is also uncertain.
- demand could exceed the available units on hand in any month, in which case all excess demand results in lost sales.
- Consequently, our production decision must find the proper balance between production costs, lost sales, and final inventory salvage value.

Stage: each of the next two months
State: inventory level at the beginning of each month
Decision: units of production at each month
Random variable: the demand $\tilde{d}_t$ for each month

Application: inventory control

Stage: each of the next two months
State: inventory level at the beginning of each month
Decision: units of production at each month
Random variable: the demand $\tilde{d}_t$ for each month

Recursion: let $f_t^*(y)$ represent the minimum cost for the remaining months with the inventory level y at the beginning of month t at stage t.
For $y = 0, 1, 2, 3$,

$$f_2^*(y) = \min_{x_2 \in \{0,1,2,3\}} \mathbf{E}_{\tilde{d}_2} \left[1000x_2 - 2000 \min\{y + x_2, \tilde{d}_2\} + (500 + 100) \max\{y + x_2 - \tilde{d}_2, 0\} \right]$$

$$f_1^*(0) = \min_{x_1 \in \{0,1,2,3\}} \mathbf{E}_{\tilde{d}_1} \left[1000x_1 - 2000 \min\{x_1, \tilde{d}_1\} + 100 \max\{x_1 - \tilde{d}_1, 0\} + f_2^*(\max\{x_1 - \tilde{d}_1, 0\}) \right]$$

Reading materials

Section 11.1-11.3, *Introduction to Operations Research* by Hillier and Lieberman